

ZAALI Mohamed

AI Agent Developer

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EDUCATION

ENSAM Rabat – State Engineer Student in Data Science, AI & Digital Health

Sep 2023–Present

PROFESSIONAL EXPERIENCE

Syntax - RABAT - AI Agent Developer Intern

FEB 2026 - Present

Building an extensible MCP server connecting AI agents to Microsoft 365 via Microsoft Graph API.

- Architected a **preset system** controlling exactly what tools an AI agent sees per context; shipped the first production preset with **55+ tools** covering SharePoint & Excel via Microsoft Graph API.
- Built a **5-step code generation pipeline** that turns Microsoft's 45MB OpenAPI spec into typed, validated MCP tools automatically adding a new tool requires a single JSON line.
- Designed the server for **zero-effort extensibility**: onboarding a new M365 service (Outlook, Teams, Calendar) requires only a JSON config, no code changes.

Technologies: Python TypeScript FastMCP LangGraph OpenAI / Azure Node.js

Al Barid Bank - RABAT - Full Stack Developer Intern

JUL 2024 - SEP 2024

Design and development of a platform for analyzing POS, SoftPOS, and e-commerce transactions.

- Built a **secure REST API** (Spring Boot, JWT) enabling fast access to aggregated transaction data.
- Designed and delivered a **responsive website** (Angular, ngx-charts) for transaction visualization and exploration.
- Created the **software architecture in UML** to ensure system maintainability and scalability.

Ministry of Youth, Culture and Communication - RABAT - Data Science Intern

JUL 2023 - SEP 2023

Development of a predictive analytics system for cultural heritage preservation.

- Developed a **predictive model** to forecast cultural heritage degradation.
- Integrated the model into a **web application** (Flask) to support experts in decision-making.
- Deployed the solution on an **Ubuntu server** (Gunicorn, Nginx), ensuring availability and performance.

PROJECTS & ACCOMPLISHMENTS

- **AL-MUQARRIR - Automatic Transcription & Meeting Minutes Generation**: Web platform for **real-time transcription**, speaker **diarization**, and automatic generation of **meeting minutes** with **semantic analysis**. | Technologies: Whisper (OpenAI) PyAnnote Gemini API Flask Angular SQLite
- **Cura - Disease & Medication Tracking**: Web application for treatment management with AI-powered **drug interaction detection** and generation of patient-friendly alerts. | Technologies: PyTorch Gemini API Flask SQLAlchemy Bootstrap SQLite
- **LipReader - Lip-Reading Application**: Application converting lip movements into text. | Technologies: TensorFlow Streamlit
- **NER - Named Entity Recognition Models**: Developed an application capable of identifying named entities. | Technologies: Python Streamlit Keras SpaCy Stanza
- **Real-Time Network Intrusion Detection System**: System capturing and analyzing network packets in real-time. | Technologies: Python Kafka PyShark Pandas Scikit-learn Joblib
- **Anxiety Level Prediction**: Platform for assessing and predicting anxiety levels based on psychological, demographic, and behavioral information. | Technologies: Python Scikit-learn Flask Bootstrap Joblib
- **MySQL-to-Hadoop Pipeline**: Designed and implemented a data migration pipeline from a MySQL database to HDFS via Sqoop, with analysis using Hive. | Technologies: Apache Sqoop Hadoop HDFS Apache Hive MySQL Docker Linux Shell
- **BigMart Sales Analysis Dashboard in Google Looker**: Retail analytics dashboard providing strategic insights on sales and inventory optimization. | Technologies: Google Looker Python

[View Dashboard](#)

TECHNICAL SKILLS

Programming Languages: Python Java scala

AI/ML: scikitlearn, PyTorch, TensorFlow, NLP

Databases: MySQL Oracle MongoDB SQLite PostgreSQL

Big Data: Hadoop Hive Sqoop Kafka Spark

Data Integration: Talend Data Warehousing ETL

Visualization & BI: Power BI Google Looker

Web Technologies & Frameworks: HTML CSS JavaScript Bootstrap Angular Streamlit Flask Spring Boot

EXTRACURRICULAR ACTIVITIES

Learning Experience Architect - Fatal Error Club ENSAM Rabat

SEP 2024 - JUN 2025

- Led workshops in data science, machine learning, and software development; co-created innovative programs.
- Optimized training outcomes through continuous evaluation and refinement.

CERTIFICATIONS

- CCNA: Introduction to Networks
 - Data Structures and Algorithms in Python
- Introduction to Data
 - Python Programming Fundamentals

LANGUAGES

- English: Fluent
- French: Fluent
- Arabic: Native